

SUBODH KUMAR

ML Engineer | AI Backend Developer | RAG & NLP Systems

Greater Noida, U.P., India | +91-7007456072 | subodh.24fd@gmail.com

Linkedin: [linkedin.com/in/er-subodh-kumar](https://www.linkedin.com/in/er-subodh-kumar) | GitHub: github.com/ksubodh9

ML Engineer with 4.5+ years of backend engineering experience, specialising in production-ready AI systems, RAG pipelines, predictive analytics, and ML model integration. Strong foundation in Python, FastAPI, and scalable backend architecture — enabling end-to-end delivery of AI products from model development to deployment. Hands-on experience in fintech and AI domains with a focus on reliability, scalability, and real-world impact.

TECHNICAL SKILLS

ML & AI	Scikit-learn, XGBoost, Random Forest, Gradient Boosting, ARIMA/SARIMA, Predictive Analytics, Feature Engineering, Model Evaluation, Time-Series Forecasting, EDA
Generative AI / NLP	RAG Pipelines, FAISS, Chroma, Vector Databases, Semantic Search, Embeddings, Prompt Engineering, Conversational AI, LLM API Integration
Backend & APIs	Python, FastAPI, PHP (Laravel), REST APIs, JWT Authentication, Swagger UI / OpenAPI, API Design & Model Serving
Deployment & Tools	Docker, Git, Streamlit, AWS S3, Hugging Face Spaces, Vercel, Supabase, Linux (CentOS), Windows Server
Databases	PostgreSQL, MySQL, MSSQL, MariaDB
Visualization	Matplotlib, Seaborn, Power BI (Basic)

PROFESSIONAL EXPERIENCE

Vayam Technologies Ltd. Aug 2023 – Present

Product Engineer / ML Integration Engineer | MaddoxPay Fintech Platform

Applied AI / ML Integration

- ▶ Integrated predictive ML models into internal dashboards for anomaly detection and operational insights, improving fraud detection accuracy.
- ▶ Built and deployed backend APIs for ML model serving and prediction workflows with reliability-focused fallback handling.
- ▶ Supported AI-enabled decision workflows through predictive analytics and backend automation, reducing manual review overhead.

Backend Engineering & Fintech Systems

- ▶ Designed and maintained high-volume recharge and bill-payment systems handling 100K+ monthly transactions.
- ▶ Implemented automated payout systems (₹1M+ daily) using Axis Bank integrations — NEFT, RTGS, and IMPS.
- ▶ Optimised database schemas and backend workflows, improving API response performance by ~2x.
- ▶ Built analytics dashboards and reconciliation systems for operational monitoring and risk analysis.
- ▶ Implemented MFA, geofencing, biometric eKYC integrations, and fraud prevention workflows.

Flyde Digital Solutions Jul 2022 – Aug 2023

Fullstack Engineer

- ▶ Delivered backend-focused full-stack applications with scalable API integrations (Razorpay, PhonePe, CC Avenue, Shiprocket, DTDC).
- ▶ Worked on AWS S3-based media storage systems and improved backend performance through query optimisation and modular development.

Saatirmind Software Services May 2021 – Jul 2022

Laravel Backend Developer

- ▶ Developed modular web applications using Laravel and MySQL; worked on backend APIs, database optimisation, and scalable workflows.

PROJECTS

DocIntel — Document Intelligence Platform

Tech: FastAPI, Python, React, RAG, Chroma, Supabase (PostgreSQL + pgvector), Gemini LLM, Docker, Hugging Face Spaces, Vercel, JWT

- ▶ Built an AI-powered document intelligence platform that transforms how users engage with complex documents through automated classification, summarisation, and citation-backed conversational retrieval.
- ▶ Engineered a Retrieval-Augmented Generation (RAG) chat pipeline using Chroma vector store and Supabase pgvector embeddings, delivering accurate, citation-backed answers grounded in document content.
- ▶ Designed an async FastAPI backend with multi-provider LLM integration (Gemini and others), JWT-based authentication, OpenAPI docs, automated document classification, intelligent summarisation, and multi-format (PDF and more) ingestion.
- ▶ Deployed a Dockerised, microservices-ready service on Hugging Face Spaces (16GB RAM, 2 vCPU) with a React frontend on Vercel; built an ephemeral-resilient design backed by a persistent database layer with proper CORS handling and health checks.

Smart AI Assistant — RAG-Powered Conversational System

Tech: Python, FastAPI, FAISS, Vector DB, RAG, LLM APIs, Laravel

- ▶ Designed and built an end-to-end RAG pipeline including data collection, embedding generation, FAISS vector indexing, and context-aware response generation.
- ▶ Implemented semantic similarity search for efficient knowledge retrieval; integrated bilingual conversational support and file attachment processing.
- ▶ Handled prompt orchestration and retrieval optimisation for scalable AI backend integration.

Career Guidance Engine API — AI-Powered Recommendation Platform

Tech: FastAPI, Python, RAG, FAISS, Vector DB, NLP, JWT, Voice APIs

- ▶ Built a personalised AI career guidance platform with custom RAG pipeline for context-aware responses and semantic search.
- ▶ Integrated FAISS-based vector retrieval and voice assistant APIs for voice-enabled conversational interactions.
- ▶ Designed scalable FastAPI backend with JWT authentication and modular architecture.

Industrial Equipment Monitoring & Predictive Maintenance System

Tech: FastAPI, Python, Scikit-learn, Streamlit, Docker

- ▶ Engineered an AI-powered predictive maintenance platform; built preprocessing and feature engineering pipelines for sensor-based industrial data.
- ▶ Implemented and compared Random Forest, XGBoost, Logistic Regression models for fault classification and failure detection.
- ▶ Developed FastAPI backend services and Streamlit dashboards for real-time monitoring; structured Docker-ready modular architecture.

Customer Churn Prediction Platform

Tech: Python, Scikit-learn, XGBoost, Pandas, NumPy

- ▶ Built an end-to-end churn prediction system; trained and evaluated Logistic Regression, Random Forest, Gradient Boosting, AdaBoost, and XGBoost models.
- ▶ Achieved 80.3% accuracy using Logistic Regression after comparative evaluation; built reusable ML pipelines for preprocessing and prediction.

Walmart Sales Forecasting System

Tech: Python, Pandas, Scikit-learn, ARIMA/SARIMA, Matplotlib

- ▶ Developed retail sales forecasting using regression and time-series techniques; implemented trend, seasonality, and forecasting analysis.

Multimodal Biometric Authentication System

Tech: Python, OpenCV, Computer Vision

- ▶ Built a biometric authentication system using OpenCV; implemented image preprocessing, feature extraction, and biometric verification workflows.

EDUCATION & CERTIFICATIONS

Executive PG Certification — Data Science & Artificial Intelligence Intellipaat	2025
Microsoft SQL Certification Intellipaat	2025
B.Tech — Information Technology Dr. A.P.J. Abdul Kalam Technical University, Lucknow	2020
Virtual Lab Development Program (Bootathon) — IIT Bombay, 2018	